

**SSG OFFICE ASSISTANT APP: A DIGITAL MANAGEMENT SYSTEM
FOR STUDENT GOVERNMENT OPERATIONS AT CEBU TECHNOLOGICAL
UNIVERSITY DAANBANTAYAN CAMPUS**

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CEBU TECHNOLOGICAL UNIVERSITY

Daanbantayan Campus

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**SSG OFFICE ASSISTANT APP: A DIGITAL MANAGEMENT SYSTEM FOR
STUDENT GOVERNMENT OPERATIONS AT CEBU TECHNOLOGICAL
UNIVERSITY DAANBANTAYAN CAMPUS**

A Capstone Project Presented to
Information Technology Department

Cebu Technological University – Daanbantayan Campus

In Partial Fulfillment of the Requirements
for the Degree Bachelor of Science in Information Technology

By

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ABSTRACT

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INTRODUCTION

Rationale of the Project

Quality education relies on effective communication and organized systems, yet student government operations at Daanbantayan Campus remain fragmented across paper-based processes and disconnected digital tools. Important announcements reach only limited group chats, leaving faculty uninformed, while membership, inventory, and financial records are stored in separate spreadsheets. Approved resolutions are also difficult to locate. These inefficiencies reduce transparency, delay services, and limit the student government's ability to serve its constituency, especially as students increasingly expect seamless digital interactions.

The challenges faced by the SSG at Daanbantayan Campus reflect broader patterns documented in educational research. A multivocal literature review on digital transformation initiatives in higher education institutions found that approximately 56 percent of institutions launch digital transformation initiatives that remain isolated rather than integrated into a coherent institutional plan **(Fernández, Gómez, Binjaku, & Meçe, 2023)**. This fragmentation problem where separate tools are used for separate functions without integration directly mirrors the situation at Daanbantayan Campus,

where the SSG uses different disconnected tools for membership tracking, financial records, and announcements.

Furthermore, the expectations of modern students regarding digital services have been documented as a key driver of institutional change. A study on the future of higher education in the era of digital transformation argued that contemporary students expect technology-rich, fully digital experiences in all aspects of their educational journey **(Akour & Alenezi, 2022)**. Students who are accustomed to instant communication and real-time updates through social media and e-commerce platforms find paper-based processes and disconnected digital tools frustrating and inefficient. The current SSG system, which relies on closed group chats for announcements and spreadsheets for records, fails to meet these expectations.

The absence of a centralized digital system also creates accountability and transparency issues. A qualitative study on graduate student perceptions of digital governance in higher education found that students recognize the potential of digital governance to enhance efficiency, decision-making, and inclusivity **(Doğan & Arslan, 2025)**. However, the research also identified critical challenges including inadequate infrastructure, inconsistent implementation, and lack of formal training challenges that directly parallel the fragmented communication and record-keeping issues at Daanbantayan Campus. Without a unified digital platform, financial transactions lack proper audit trails, legislative documents become difficult to locate, and faculty members remain excluded from important campus announcements.

Digital records management has been shown to address these exact problems. A study on digital transformation in records management at a private higher education institution in the Philippines found that digital records management significantly reduces document loss, improves retrieval times, and enhances data accuracy (**Wagan, Briones, Baldovino, & Refozar, 2025**). The research documented that paper-based systems create significant inefficiencies, including delayed retrieval times and increased security risks. These findings directly support the development of the SSG Office Assistant App, which aims to replace fragmented paper-based tracking with a centralized digital database.

The SSG Office Assistant App addresses these challenges by providing a unified platform where member records, financial documents, product inventory, and announcements are consolidated into a single, accessible system. Role-based access control ensures that each officer interacts only with features relevant to their responsibilities. Real-time data synchronization ensures that all users see current information. Automated chatbot integration bridges the communication gap between students and faculty. Through these features, the project delivers a scalable digital hub that transforms student government operations from a fragmented, reactive system into an integrated, proactive platform for student governance.

Review of related Systems/Technical Background

This section presents a review of existing digital platforms and related studies to understand how similar organizational problems have been addressed. The review links the SSG Office Assistant App to established digital approaches, ensuring the proposed system addresses information gaps caused by fragmented, paper-based processes. By examining these models, the study provides a clear technical direction for developing a more transparent and efficient student government system at Daanbantayan Campus.

The shift from paper-based to digital records management in educational institutions has been widely documented. Studies show that digital systems reduce document loss, improve retrieval time, and enhance data accuracy, while manual processes lead to delays, duplication, and security risks **(Wagan, Briones, Baldovino, & Refozar, 2025; Nweke, 2025)**. These findings support the development of the SSG Office Assistant App, which replaces fragmented tracking with a centralized digital database. The system incorporates key document management features such as metadata tracking, search functionality, and access controls, allowing Legislative Branch members to upload resolutions with details like date created, authors, tags, and committee, and retrieve them efficiently.

Digital governance in higher education offers improved efficiency and decision-making, but institutions often face challenges such as limited awareness, inadequate infrastructure, inconsistent implementation, and lack of training **(Doğan & Arslan, 2025)**. In addition, many digital transformation efforts remain fragmented, with systems implemented separately rather than as part of an integrated solution **(Fernández, Gómez, Binjaku, & Meçe, 2023)**. These issues parallel the situation at Daanbantayan Campus, where communication gaps and records are spread across multiple spreadsheets. The SSG Office Assistant App addresses this fragmentation by consolidating member management, inventory, financial tracking, document management, and announcements into a single, integrated dashboard.

Digital tools have been shown to streamline administrative operations in higher education. A comprehensive review of digital transformation in the management of higher education institutions examined how digital tools streamline operations, automate routine processes, and improve overall performance **(Carmo, Lacerda, Klingenberg, & Piran, 2025)**. The study established that digital transformation enables institutions to replace manual, time-consuming administrative tasks with automated workflows that reduce errors and improve response times. This finding supports the SSG Office Assistant App's approach of automating key student government functions, including order processing, financial tracking, and announcement distribution.

Student expectations are driving digital transformation in education. A study on the future of higher education in the era of digital transformation argued that contemporary students expect technology-rich, fully digital experiences in all aspects of

their educational journey **(Akour & Alenezi, 2022)**. The research established that students who experience seamless digital interactions in other areas of their lives increasingly demand the same level of convenience, speed, and transparency from educational administration and student services. This finding supports the rationale for the SSG Office Assistant App, which addresses the expectations of modern students at Daanbantayan Campus.

Security of sensitive educational data is a key concern in digital transformation. Studies show that Role-Based Access Control (RBAC) improves system security while maintaining operational efficiency by limiting access based on user responsibilities **(Liu et al., 2017; Sabo, Umaru, Mohammed, & Yusuf, 2025)**. Field-tested school management systems using RBAC also demonstrated better protection against unauthorized access and improved system performance. These findings support the SSG Office Assistant App's implementation of RBAC, where users are classified into eight distinct roles with specific permissions to protect sensitive financial and membership data.

Web-based university management systems have also shown significant benefits. A study on a university management system that integrated role-based access control, centralized record management, and data visualization capabilities demonstrated that such integrated systems significantly reduce administrative workload **(Mehta & Paneri, 2025)**. The research found that eliminating redundant data entry and providing unified access to information across different functional areas improves efficiency. This finding is directly relevant to the SSG Office Assistant App, which similarly integrates multiple functions into a single, role-based dashboard.

In conclusion, the literature demonstrates that digital management systems are not merely storage tools they are dynamic platforms that streamline operations, enhance transparency, and improve communication. Manual document management increases risks of loss and unauthorized access **(vmedulife, 2025; Nweke, 2025)**. Digital governance faces challenges including inadequate infrastructure and lack of training **(Doğan & Arslan, 2025)**. Many institutions have fragmented, non-integrated systems **(Fernández et al., 2023)**, and modern students expect digital experiences **(Akour & Alenezi, 2022; Hashim et al., 2021)**. Role-based access control is essential for protecting sensitive data **(Liu et al., 2017; Sabo et al., 2025)**. Building on these foundations, the proposed SSG Office Assistant App integrates repository and workflow capabilities to ensure that student government operations are systematically managed, transparent, and accountable through a unified digital platform.

Technical Background

The development of the SSG Office Assistant App is anchored on a separated client-server architecture designed to replace fragmented, paper-based workflows with a high-availability digital ecosystem. By leveraging React and Vite for the frontend, the system ensures a responsive, component-based user interface that minimizes latency during administrative tasks. This is paired with Firebase, a serverless backend-as-a-service (BaaS), which provides a NoSQL Firestore database for real-time data synchronization and Firebase Authentication for robust role-based access control. Unlike traditional relational databases, this stack is chosen for its elasticity allowing the student government to manage fluctuating volumes of member records, financial data, and

inventory without the overhead of local server maintenance. The integration of the Facebook Messenger API further extends the system's reach, bridging the communication gap between student leaders and faculty through automated, cross-platform notifications.

Front End Stack

The frontend strategy utilizes a component-driven architecture powered by React and Vite to transform complex administrative tasks into a modular and responsive user interface. By decomposing the dashboard into independent, reusable functional units, the system ensures consistent performance across diverse user roles while significantly reducing maintenance overhead. This approach facilitates a Single Page Application (SPA) environment that eliminates traditional page reloads, providing student leaders with the speed and fluid navigation required for modern digital governance.

Data Strategy

The data strategy centers on Firebase Firestore, a cloud-native, NoSQL document-oriented database that facilitates real-time data persistence and synchronization through a serverless architecture. This schema-less model is prioritized over traditional relational databases because it allows for a flexible data structure, enabling the system to store diverse records such as hierarchical member profiles, financial receipts, and legislative metadata without the overhead of rigid table migrations. The significance of this strategy lies in its ability to eliminate information silos by providing a single, reactive source of truth that ensures all departments have instant access to synchronized data, which is critical for maintaining transparency in student governance.

Security Strategy

The security strategy implements Role-Based Access Control (RBAC) leveraging Firebase Authentication and JSON Web Token (JWT) Custom Claims to enforce technical permission levels across the platform. This technical approach assigns specific administrative "claims" to authenticated identities such as Budget Sector, Creative Team, or President which are validated at both the interface and the database layer. To ensure backend integrity, Firestore Security Rules programmatically intercept every incoming request, verifying the user's cryptographic token before permitting any CRUD (Create, Read, Update, Delete) operations. This architecture is significant because it establishes a Zero-Trust environment, ensuring that sensitive financial ledgers and student records remain isolated from unauthorized users while providing a clear audit trail for administrative accountability.

Connectivity Strategy

The connectivity strategy employs an asynchronous event-driven architecture via Webhook Integration with the Facebook Messenger API. This technical implementation utilizes automated server-side triggers that execute whenever a high-priority event such as a legislative resolution or a campus-wide announcement---is committed to the cloud database. By pushing a data payload to external API endpoints, the system facilitates cross-platform synchronization, broadcasting real-time notifications to faculty and staff within their native communication environments. The significance of this strategy is the establishment of a proactive outreach layer that eliminates information silos, ensuring

that stakeholders remain informed through automated distribution rather than relying on manual, fragmented messaging.

THE PROBLEM

Statement of the problem/Objectives of the project

The study aims to analyze, design, and implement the SSG Office Assistant App to enhance administrative efficiency, communication transparency, and service delivery for the Student Government Organization and the academic community at Daanbantayan Campus.

Specifically, it seeks to answer the following questions:

1. What is the current status of the Student Government Organization's operations in terms of:
 - 1.1 Student membership and organization record management;
 - 1.2. Financial document tracking and resolution accessibility;
 - 1.3. Information dissemination and faculty-student communication; and
 - 1.4. Inventory and office resource monitoring?
2. What are the challenges encountered with the current manual and fragmented system in managing student government processes?
3. Based on the aforementioned services, to what extent is the effectiveness of the SSG Office Assistant App in terms of:
 - 3.1. Role-based user account management and security;
 - 3.2. Centralized data organization and record accuracy; and
 - 3.3. Automated communication and information dissemination (e.g., via Messenger Chatbot)?

4. To what extent does the SSG Office Assistant App improve the efficiency of student government operations?
5. To what extent is the system acceptable to users In terms of :
 - 5.1. Usability;
 - 5.2. Functionality;
 - 5.3. Reliability; and
 - 5.4. Performance?

Hypothesis

This study focuses the objectives that determine whether *SSG Office Assistant App: A Digital Management System For Student Government Operations* contributes to proper efficiency and improvement to traditional student government local service operations from storage management, record keeping and centralized communication for those benefited by both students and faculties at Cebu Technological University – Daanbantayan Campus.

To guide the analysis, particularly the comparison between the challenges of the existing process and the effectiveness of the developed system, the study utilizes the following hypothesis.

Null Hypothesis (H_0): There is no significant difference between the challenges encountered in the current traditional record keeping and student government process and the perceived effectiveness of a developed centralized digital student government operation.

Alternative Hypothesis (H_1): There is a significant difference between the challenges encountered in the current traditional record keeping and student government process and the perceived effectiveness of a developed centralized digital student government operation..

Scope and Limitations of the project

This project focuses on developing a React-based administrative dashboard application designed to streamline and modernize student government operations at Cebu Technological University–Daanbantayan Campus. The system supports the digital management of key organizational functions, including member records, product inventory, order processing, financial tracking, document handling, and official announcements. By integrating these processes into a single centralized platform, the application aims to improve efficiency, reduce reliance on manual and fragmented systems, and enhance the accuracy, accessibility, and transparency of data.

User Authentication and Access Control. The system encompasses user authentication and access control through Firebase Authentication, implementing role-based access for General Students, Creative Team, House Representatives, Governors, Legislative Branch, President, Budget Sector, and System Administrators, while allowing public access to announcements, calendar events, and product catalog without requiring login.

Member Management. The member management feature enables authorized users to add, view, edit, and delete student member profiles with photographs, including department-based filtering for House Representatives and Governors to view members specific to their department such as BSIT or BEED.

Product Inventory and Order Processing. The product inventory and order processing module allows Legislative Branch members to manage the product catalog,

while students can submit orders after login, track their order status, and view order history, with department-specific order views available for Representatives and Governors.

Finance Management. The finance management module provides budget tracking for income and expenses with receipt attachment capabilities, restricted exclusively to the President and Budget Sector.

Document Management. The document management system allows Legislative Branch members to upload and store approved legislative documents with metadata tracking including date created, authors, tags, and committee, featuring search and filter functionality.

Announcement and Information Dissemination. For information dissemination, the Creative Team manages announcements and calendar events, displaying a "What's New" feed for students, with Facebook Chatbot integration enabling automated distribution of announcements to subscribed teachers and faculty via Messenger.

User Interface. The user interface features a responsive design with role-based dashboards and consistent header and sidebar navigation.

Limitations

The study acknowledges several limitations. The application is designed specifically for the Student Government Organization at Daanbantayan Campus and may not suit other institutions without significant modifications. It requires an active internet connection with no offline functionality, and relies on third-party services including Firebase and Facebook API, making it subject to external policy changes. The system

focuses only on SSG operations and does not function as a comprehensive enterprise resource planning system for institutional management. It does not integrate with payment gateways, so actual payments must be handled offline. Receipt and document uploads are limited by Firebase storage capacity. Finally, the effectiveness of the system depends on user acceptance and consistent usage across all stakeholders.

Significance of the study and project highlights

The Supreme Student Government often remains tethered to inefficient, manual process. While students are the primary adopters of modern technology, the system they use to lead frequently rely on fragile paper-based workflows and fragmented communication channel. This creates a *"digital lag"* that hinders the growth of the student government and delays the delivery of essential services to the campus community.

The development of the SSG Office Assistant App serves as a vital step toward digital modernization of student governance at Daanbantayan Campus. The study is significant to the following:

For the Student Body:

The students are the ultimate beneficiaries of improved governance. Through this system, students gain a higher level of transparency regarding how university funds are utilized and how legislative decisions are made. Faster service delivery and easily searchable documents ensure that student needs are met with efficiency and fairness.

For the University Faculty and Administration:

This study addresses the long-standing "information gap" where faculty members were often the last to know about campus events. By integrating a more inclusive dissemination system and an automated chatbot, the project fosters a more collaborative environment between the student body and the university staff, ensuring administrative oversight is informed and timely.

For Future Researchers:

This study provides a technical and conceptual foundation for future developers exploring React-based administrative dashboards and Firebase integration in an academic setting. It offers documented insights into the challenges of digital governance and the efficacy of automated communication tools in improving institutional transparency.

For the Supreme Student Government (SSG) Officers:

This research provides student leaders with a centralized tool to eliminate the *"administrative chaos"* of paper-based tracking. By automating inventory and record-keeping, officers can reduce manual errors, save time, and focus more on student-centric initiatives rather than tedious paperwork. The role-based access ensures that each department---from the Budget Sector to the Creative Team---can operate with clarity and accountability.

METHODOLOGY

The methodology for developing the SSG Office Assistant App: A Digital Management System for Student Government Operations at Cebu Technological University – Daanbantayan Campus employed a developmental research progression and utilized the System Development Life Cycle (SDLC) following the Agile model. Agile adopts an iterative and incremental process where development occurs in short cycles called sprints. Each sprint integrates planning, development, testing, and user feedback,. This flexible approach was particularly suited to addressing the dynamic operational needs identified during the Needs Assessment phase. The section of this study discusses the methods and procedures, including the study's locale and population, data instruments, research design, research procedures, data gathering methods, structured data analysis, statistical treatments, and the techniques used for data collection and interpretation. The methodology also includes the technical design and the project's environmental specifications, ensuring that the platform's architecture, workflows, and interface support efficient operation management, improved accessibility, and a more systematic review process for both students and faculty.

Environment

Locale

Figure 1 presents the location of Cebu Technological University–Daanbantayan Campus situated in Barangay Agujo, northern Cebu. The campus provides various undergraduate programs, including Fisheries, Industrial Technology, Engineering, Information Technology, Hospitality Management, and Teacher Education. It is located

under the jurisdiction of the Municipality of Daanbantayan, Cebu, which belongs to Region VII in the Central Visayas.

Cebu Technological University–Daanbantayan Campus serves as a vital educational institution within the municipality, with a strong focus on undergraduate education, research development, and community-oriented academic programs. The campus provides essential facilities Supreme Student Government Office that aim to address and respond to student needs. It also implements various academic programs and institutional policies designed to promote student welfare and development.

In Addition, The Supreme Student Government Office offers a range of extracurricular activities that encourage student engagement beyond the classroom. These activities help students develop personal skills and social responsibility that contributes to a more collaborative and inclusive campus environment. However, the Student Government Office continues to face significant challenges due to its reliance on traditional paper-based record keeping. Important documents such as financial receipts, policy records, and official reports are manually stored and maintained, which makes the system inefficient and prone to disorganization.

As the volume of records increases over time, location and retrieving specific information becomes more difficult and time-consuming. This issue also extends to the management of student-related assets, particularly campus merchandise such as T-shirts, uniforms, and lanyards. Since records for these items are often recorded in physical logs or separate spreadsheets, tracking purchases, inventory levels, and distribution history becomes inconsistent and prone to errors. As a result, data may become duplicated,

misplaced, or outdated, further affecting transparency and accountability in student government operations.

The focus of this study is on the campus as the primary location for implementing and Supreme Student Government Assistant App ensuring the systems aligns with the institution's infrastructure, academic practices, and local context.

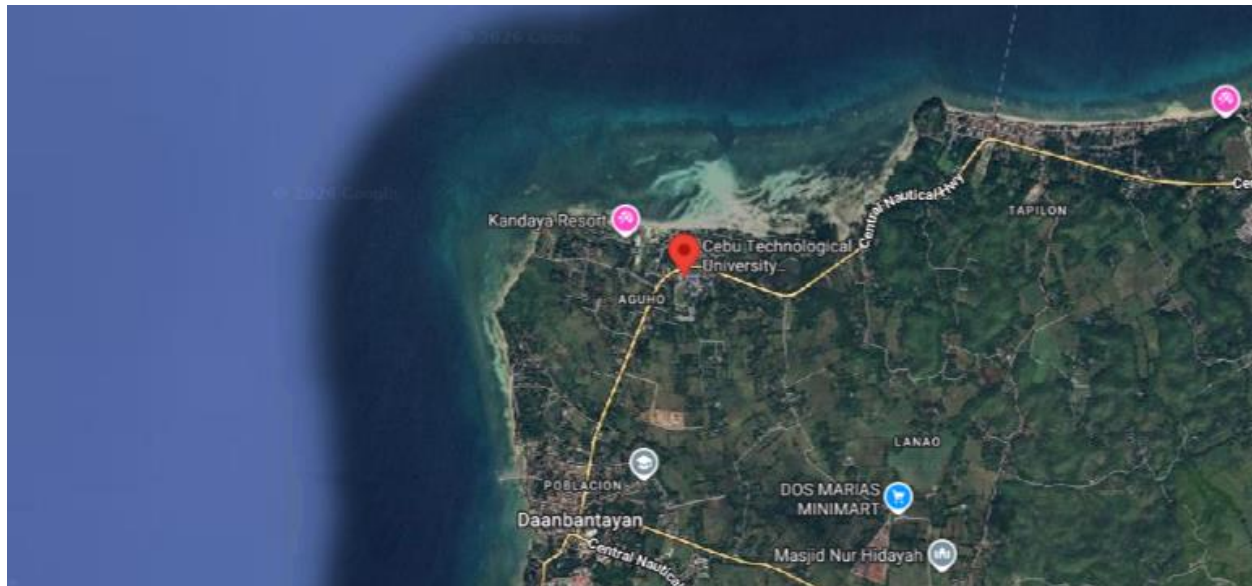


Fig1. Location of Cebu Technological University- Daanbantayan Campus

Population of The Study

The population of the study consisted of undergraduate students from Cebu Technological University – Daanbantayan Campus who are actively engage (need more data gathering

Data Instruments

The primary data-gathering instrument used in this study was a structured survey questionnaire designed to assess the operational challenges, administrative bottlenecks, and satisfaction levels of SSG officers and students regarding the current paper-based record-keeping system and the proposed SSG Office Assistant App. The questionnaire included statements related to data retrieval speed, record accuracy, accountability in financial tracking, and the overall reduction of student wait times compared to traditional manual folder management. Respondents evaluated each statement using a 5-point Likert scale, complemented by open-ended sections for officers to provide qualitative observations on recurring administrative friction points.

Table 1. Numerical Scale

Scale	Descriptive Equivalent
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

Research Design

This study utilizes a descriptive-developmental research design to evaluate current office management practices and assess the effectiveness of the developed SSG Office Assistant App. The descriptive design allows the researchers to document existing

bottleneck in manual folder tracking, including data redundancy, slow record retrieval, and physical storage risks. The developmental component focuses on the implementation of a cloud-native, real-time architecture to ensure data integrity and operational transparency. The design also utilizes a quantitative evaluation to determine if the digital transition significantly reduces student wait times and improves administrative accountability at the Daanbantayan Campus.

Research Procedures

The research followed a systematic three-phase procedure. The first phase, Needs Assessment, employed a mixed-method survey distributed to SSG officers, students, and faculty to identify operational bottlenecks within the existing paper-based system. Documented issues—including data redundancy, slow retrieval times, and information silos—were then translated into a formal User Requirements Specification. In the second phase, System Development, an Agile methodology was used to construct a digital application architecture designed to resolve these identified gaps. The third phase, System Evaluation, utilized a quantitative validation instrument based on a 5-point Likert scale to measure the application's performance against prior manual methods. The results of these procedures formed the basis for analyzing the application's effectiveness in reducing student wait times, bridging faculty communication gaps, and enhancing administrative accountability at the Daanbantayan Campus.

Data Gathering Methods

Data were gathered using a mixed-method approach, combining online surveys distributed through digital platforms (e.g., Google Forms) and physical questionnaires administered on campus, depending on the accessibility and convenience of the SSG officers, students, and faculty respondents. Clear instructions were provided to ensure accurate and honest responses. The questionnaire focused on the challenges encountered in the existing paper-based system—specifically operational bottlenecks such as data redundancy, slow folder retrieval, and the "information silo" effect that prevents timely updates between the SSG office and school authorities. Additionally, respondents were asked about their perceptions of the expected effectiveness and usability of the proposed digital management system, including features like real-time data synchronization and automated faculty notifications.

Data Analysis

The collected data were coded, tabulated, and subjected to descriptive statistical analysis, including frequency counts, percentages, and means. These measures were used to summarize respondent profiles and address Problem 1 (the current status of SGO operations in terms of membership and record management, financial document tracking, information dissemination, and inventory monitoring) as well as Problem 2 (the challenges encountered with the current manual and fragmented system). For inferential analysis, a paired-sample t-test was employed to compare respondents' perceptions of the existing manual system (addressing Problems 1 and 2) against their perceptions of the developed SSG Office Assistant App (addressing Problems 3, 4, and 5: effectiveness,

efficiency, and acceptability in terms of usability, functionality, reliability, and performance). This approach allowed the researchers to determine whether the digital system provided statistically significant improvements in administrative efficiency, communication transparency, and service delivery at the Daanbantayan Campus.

Statistical Treatment

The study utilized weighted mean and standard deviation to analyze respondent perceptions, while a paired-sample t-test determined if the SSG Office Assistant App provided a statistically significant improvement over the manual system in terms of efficiency, transparency, and service quality.

Weighted Mean. The weighted mean is used to measure the central tendency of respondents' perceptions by assigning numerical weights to each response option. Each response is multiplied by its assigned weight, the products are summed, and the total is divided by the number of responses.

Formula:

$$\bar{X} = \frac{\sum X}{N}$$

Where:

$\bar{X}$ = Weighted mean

$\Sigma(x)$ = Summation of the products of each weight and its corresponding score

X = Scores assigned of each respondent's perception

N = Total number of responses

Standard Deviation. The standard deviation is used to measure the dispersion or variability of responses around the mean, indicating how consistent the respondents' answers were. A low standard deviation suggests high agreement among respondents, while a high standard deviation indicates diverse opinions.

Formula:

$$\sigma = \sqrt{\frac{\sum_i^n (x_i - \bar{x})^2}{n - 1}}$$

Where:

σ = Population standard Deviation

Σ = Summation (add all values)

X = Individual response value

$\bar{x}$ = Sample mean

n = Number of Scores in the Sample

Average Weighted Mean. The Average weighted mean is a treatment used to obtain the overall perception of respondents across multiple indicators. It is computed by

averaging all weighted means generated from the questionnaire items, representing the general level of agreement or assessment.

Formula:

$$AWM = \frac{\sum(fw)}{N}$$

Where:

AWM = Average Weighted Mean

Σ = Summation

f = Frequency

w = Weight

N = Total number of respondents

Paired Sample t-Test. The paired-sample t-test is an inferential statistic used to determine whether there is a significant difference between two related sets of scores taken from the same respondents under two different conditions or time points. It compares the mean difference between paired observations against zero.

Formula:

$$t = \frac{\bar{D}}{S_D/\sqrt{n}}$$

Where:

t = Computed t-value

$\bar{D}$ = Mean of the differences between paired scores

SD = Standard deviation of the differences

N = Number of paired responses

Table 2. Scoring Procedure for the Perceived Effectiveness, Efficiency, and Acceptability of the SSG Office Assistant App

Mean Score	Interpretation	Verbal Description
4.21 – 5.00	Strongly Agree	The statement is highly applicable and strongly reflects the actual experience with the SSG Office Assistant App.
3.41 -4.20	Agree	The statement is generally applicable and aligns well with actual user experience.
2.61 – 3.40	Neutral	The statement is moderately applicable to the actual situation.
1.81 – 2.60	Disagree	The statement shows limited applicability to the actual situation.
1.00 -1.80	Strongly Disagree	The statement is not applicable to the actual situation.

Table 2, Scoring Procedure for the Perceived Effectiveness, Efficiency, and Acceptability of the SSG Office Assistant App, presents the rating scale used for interpreting the weighted mean scores derived from the respondents' answers. Scores ranging from 4.21 to 5.00 indicate "Strongly Agree," meaning the system is viewed as

highly applicable and effective. Scores from 3.41 to 4.20 reflect "Agree," representing favorable perceptions aligned with actual user experience. Weighted means within 2.61 to 3.40 are labeled "Neutral," indicating moderate applicability. Scores between 1.81 and 2.60 indicate "Disagree," showing limited perceived relevance, while scores from 1.00 to 1.80 represent "Strongly Disagree," suggesting the indicators do not reflect the respondents' actual experience. This scoring procedure provides a structured and consistent basis for interpreting the effectiveness of the developed system.

Organizational Chart (WORK IN PROGRESS – FIGURE)

The organizational structure of the SSG Office Assistant App illustrates a coordinated system of role-based authority among administrators, specialized sectors, and student stakeholders. This framework enables the effective implementation of the digital dashboard, ensuring that data management is strictly governed through distinct access levels. By aligning the system's permissions with the campus's leadership hierarchy, the platform promotes administrative accountability, secure record-keeping, and seamless communication across the Daanbantayan Campus community.

Attached Organizational Chart

Fig 2. Functional Development Supreme Student Government Chart

System Requirement

The development of the SSG Office Assistant App at Cebu Technological University – Daanbantayan Campus was evaluated based on three key feasibility dimensions: Operational, Technical, and Schedule Feasibility. Each aspect ensures that the system is practical, sustainable, and directly responsive to the administrative challenges and institutional needs of the student government.

Operational Feasibility

The SSG Office Assistant App: A Digital Management System for Student Government Operations is operationally feasible as it addresses the inefficiencies currently experienced by the Supreme Student Government (SSG) at Daanbantayan Campus, Agujo, Daanbantayan, Cebu. At present, SSG operations rely on manual processes and scattered digital tools, resulting in disorganized records, delayed services, and communication gaps among officers, members, faculty, and students. Essential documents such as membership lists, financial reports, inventory records, and resolutions are stored in multiple formats, making retrieval and management difficult. In addition, announcements are often limited to specific group chats, excluding other stakeholders such as faculty, while financial tracking and document monitoring remain decentralized, affecting transparency and accountability.

To address these issues, the proposed system introduces a centralized digital platform that consolidates key SSG functions into one system. It streamlines processes such as membership management, inventory tracking, service requests, financial monitoring, and announcements, improving accessibility and reducing data fragmentation.

The system is designed for various stakeholders, including SSG officers from different sectors, faculty advisers, and student members. Each user is assigned specific roles and access levels to ensure organized task handling and secure data management.

Overall, the system is expected to enhance operational efficiency, improve communication, strengthen transparency, and speed up service delivery. It also supports the institution's goal of digital transformation by modernizing student government operations into a more organized, accessible, and accountable system.

Fishbone Diagram

A fishbone diagram is used to identify the root causes of inefficiency in Student Government Office operations. The main problem, such as delayed services, poor communication, and disorganized records, is placed at the "head," while contributing factors branch out as categories like people, processes, technology, documents, and communication. Each category represents possible causes such as unclear role assignments, manual workflows, scattered files, lack of a centralized system, and ineffective announcement dissemination. By organizing issues in this way, the diagram

helps clearly trace operational problems back to their sources, making it easier to understand why inefficiencies occur and supporting the need for a centralized digital solution.

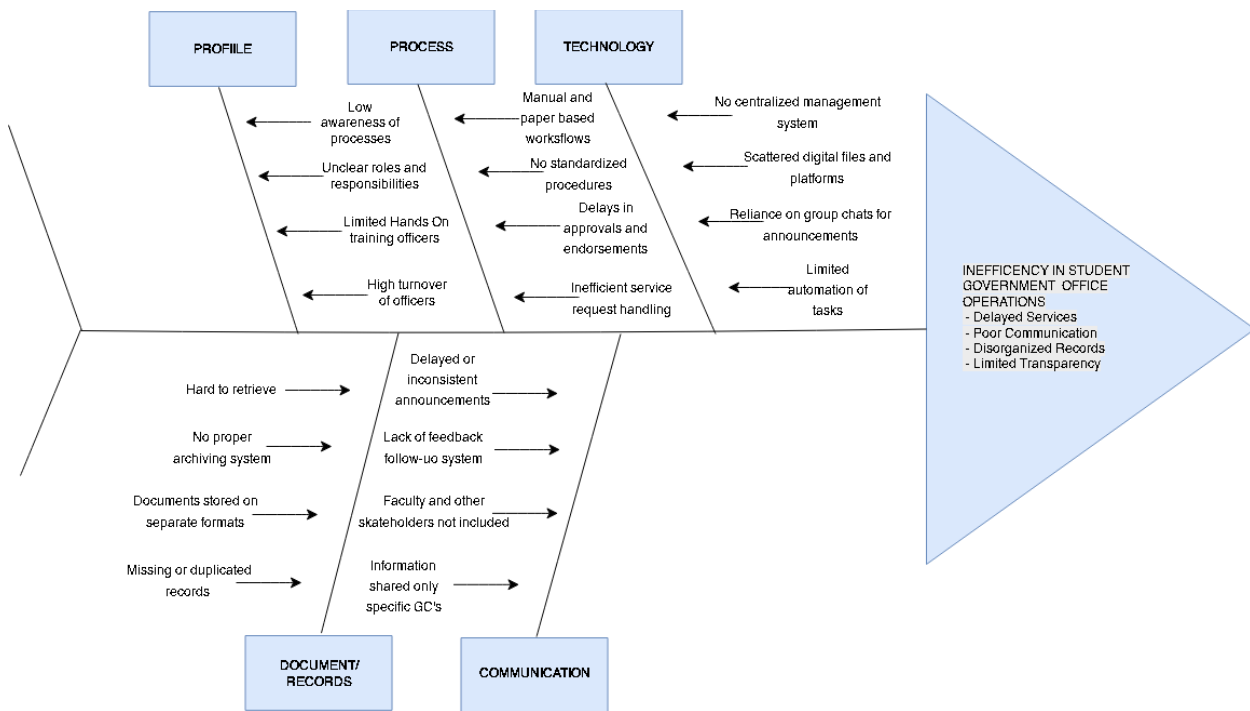


Fig 3. Fishbone Diagram – Causes of Inefficiency in Student Government Office Operation

Functional Decomposition Diagram (WORK IN PROGRESS - FIGURE)

The Functional Decomposition Diagram (Figure 4) illustrates the modular architecture of the SSG Office Assistant App, detailing how high-level operations are broken down into specific, actionable technical processes. The system is partitioned into four core engines: Identity Management for role-based authentication, Digital Archiving

for manual folder virtualization, Fiscal Tracking for real-time receipt processing, and an Automated Notification Gateway via the Messenger API.

By deconstructing these complex administrative tasks into manageable sub-functions, the architecture ensures that every user interaction—from an officer updating a budget log to a student receiving a fine notification—is executed with precision. This modular approach guarantees that the system remains scalable, secure, and highly responsive, directly addressing the critical need for a centralized, paperless infrastructure within the Daanbantayan Campus.

Fig 4. Functional Decomposition Diagram -System Components

Technical Feasibility

Technical Feasibility is an assessment of whether an organization has the necessary technological resources and expertise to develop, implement, and maintain a proposed system. It focuses on confirming that the project can be realistically built using current hardware and software without exceeding the team's technical capabilities.

The development of the SSG Office Assistant App: A Digital Management System for Student Government Operation is technically feasible due to the use of well-established open-source web frameworks like React. This frontend technology provides a component-based architecture that ensures the application is modular, easy to maintain, and highly responsive across both desktop and mobile devices.

To support this frontend, the system utilizes Firebase as a serverless backend infrastructure. This choice is technically advantageous because it provides real-time data synchronization and cloud storage without the complexity of managing physical servers. Security is further reinforced through Firebase Authentication and Role-Based Access Control (RBAC), which ensures that sensitive student records and financial data are accessible only to authorized personnel. Additionally, the integration of the Facebook Messenger API via webhooks allows the system to bridge communication gaps using existing, reliable networking protocols. Since the app operates on standard web-enabled devices, it remains a sustainable and technically viable solution for the long-term administrative needs of the campus.

Schedule Feasibility

The Schedule Feasibility of the SSG Office Assistant App was meticulously planned to align with the academic milestones set by the University's Capstone manual, ensuring all technical and research requirements were met within the semester.

From January to February, the focus was on the foundational stages, involving a deep dive into the Capstone Manual and the development of core components such as the Introduction and Scope. During this period, three potential titles were conceptualized and prepared for initial review. By March 25, the team shifted to intensive preparation for the Title Hearing, where the project was officially approved under its current title.

In April, the project transitioned into formal documentation, focusing on the completion of Chapters 1 and 2. Looking ahead, the month of May is dedicated to the development of the system prototype, including the design of functional flowcharts and technical diagrams. This phase is critical for the upcoming Design Hearing, ensuring that the system's architecture and user interface are fully prepared for final evaluation according to the prescribed research timeline.

Gantt Chart (WIP)

Insert here

Fig 5. Gantt Chart – Project Timeline

Requirements Modeling

The system requirements for the SSG Office Assistant App were modeled using both process-based and object-oriented approaches to ensure a seamless transition from manual workflows to a digital environment. These modeling techniques were employed to define the precise flow of administrative data, the logic of role-based

process controls, and the specific interactions between SSG officers and the student body.

Input-Process-Output Diagram

The Input-Process-Output (IPO) Model (Figure 6) illustrates how the SSG Office Assistant App transforms manual administrative data into a streamlined, digital management system. It outlines the flow from Inputs, such as student records, financial transaction details, and officer credentials, through Processes like role-based authentication, automated receipt generation, cloud-based data archiving, and Messenger API notifications.

The resulting Outputs include a centralized, searchable database and real-time financial transparency for the student body. This model emphasizes how systematic digital processing enhances organizational efficiency, reduces paper dependency, and ensures secure data preservation within the Cebu Technological University – Daanbantayan Campus student government operations.

Fig 6. IPO Diagram – Input-Process-Output-Flow

Process

The system architecture of the SSG Office Assistant App includes interconnected processes tailored to specific organizational roles. These processes encompass role-based authentication, task-specific access (Student, Officer, and Administrator), digital

record submission, and real-time financial tracking. Each process is meticulously designed to eliminate the redundancies of manual paper-filing and promote efficient data handling. By automating these workflows, the system ensures a streamlined administrative experience, allowing for instantaneous file retrieval and transparent communication within the student government.

Control

Control logic is strictly enforced within the SSG Office Assistant App to maintain high standards of security and data consistency. For processes such as financial collections, uploaded proofs of payment are initially marked as "pending" and require verification by the SSG Treasurer before being officially recorded in the digital ledger.

Role-Based Access Control (RBAC) is implemented to ensure that sensitive actions, such as editing student records or approving budget allocations, are restricted to authorized officers only. Additionally, the system incorporates input validation for all digital forms and leverages Firebase session management to maintain system integrity, ensuring that every administrative action is logged and reliable.

Data and Processing Modeling

To thoroughly understand the system's functionality, data and process modeling techniques were employed to capture how information moves within the SSG Office Assistant App and how key administrative processes interact. This approach helps define clear system boundaries, inputs, outputs, and controls while visualizing the flow of data among SSG officers, students, and faculty advisers.

By mapping these interactions through diagrams—such as the Functional Decomposition Diagram and system flowcharts—the development team ensures that all essential operations, including digital receipting, fiscal tracking, record archiving, and role-based authentication, are logically structured and well-coordinated. This modeling not only clarifies system behavior for campus stakeholders but also provides a strong foundation for the technical implementation and deployment within the Cebu Technological University – Daanbantayan Campus.

Context Diagram

The Context Diagram (Figure 7) illustrates the high-level interaction between the SSG Office Assistant App and its primary external entities. It visualizes how students, SSG officers, and faculty advisers exchange data with the system through processes such as payment verification, digital record submission, and notification broadcasting. This diagram provides a clear overview of how the platform centralizes student government operations, ensuring a secure, organized, and transparent administrative workflow

Fig 7. Context Diagram – System Boundary and External Entities

Data Flow Diagram

The Data Flow Diagram (Figure 8) illustrates how information moves within the SSG Office Assistant App. Students and officers interact with the system through processes such as payment submission, transaction verification, and digital record archiving, while the SSG Treasurer and Secretary manage validation, record-keeping, and information dissemination. Once verified, administrative data and financial records are securely stored in the Firebase cloud database, ensuring real-time transparency, efficient collaboration, and easy accessibility for the student government and faculty advisers

Fig 8 Data Flow Diagram

System Flowchart

The System Flowchart (Figure 9) presents the main workflow of the SSG Office Assistant App. It begins with user authentication and role verification, directing users to their specific interfaces as Students, SSG Officers, or Faculty Advisers. Students upload proofs of payment or access their student records, while SSG Officers (Treasurer and Secretary) review submissions, generate digital receipts, and manage the administrative ledger. Finally, the Faculty Adviser performs oversight and final validation of the fiscal reports. This process ensures a structured, role-based progression from initial data entry to verified digital archive.

Fig. 9 System Flowchart

Program Flowchart (highlights only)

The Program Flowchart (Figure 10) illustrates the general sequence of user interactions within the SSG Office Assistant App. The process starts with login

authentication via Firebase, followed by the dynamic loading of the dashboard based on the user's specific role. Users then perform specialized tasks, such as submitting payment proofs, reviewing student financial statuses, or managing organizational records. Every administrative activity is synchronized in real-time for accountability before the session concludes through a secure logout. This flow ensures that all user operations are structured, traceable, and strictly role-driven.

Fig. 10 Program Flowchart

Object Modeling

Object modeling provides a structured visual representation of the entities within the SSG Office Assistant App and how they interact. It focuses on identifying key objects such as Students, SSG Officers, Administrative Records, and Faculty Advisers, along with their specific attributes and behaviors. These models serve as the foundation for designing class diagrams, use case diagrams, and sequence diagrams that reflect the system's logic, user interactions, and internal administrative processes. By mapping out relationships and responsibilities—such as the link between a student's payment and the treasurer's verification—object modeling ensures the system is scalable, maintainable, and strictly aligned with the real-world organizational workflows.

Use Case Diagram

The Use Case Diagram (Figure 11) provides an overview of the primary interactions within the SSG Office Assistant App. It illustrates how various user roles—Students, SSG Officers (such as the Treasurer and Secretary), and Faculty Advisers—engage with the

system. Students utilize the platform to submit payment proofs and monitor their financial standing, while SSG Officers manage the verification of transactions, maintain organizational records, and issue digital receipts. The diagram highlights the collaborative workflow that ensures transparent fiscal management and secure data handling for the student government.

Fig. 11 Use Case Diagram

Class Diagram

Figure 12 illustrates the core structure of the SSG Office Assistant App through key classes such as *User*, *FinancialRecord*, *Officer*, and *DigitalReceipt*. It defines essential attributes and methods for handling tasks such as Firebase authentication, payment submission, transaction verification, and the generation of digital ledgers.

Specialized classes like *Student* and *Officer* (including specific roles for the Treasurer and Secretary) inherit from the base *User* class, while a dedicated *AccessControl* class ensures that role-based permissions are strictly enforced. This diagram forms the foundation for the system's Firebase database schema and object-oriented logic, providing a scalable and organized digital management framework

Fig. 12 Class Diagram

Sequence Diagram

Figure 13 shows the Sequence Diagram of the SSG Office Assistant App, illustrating the step-by-step interaction among the student, the system, SSG officers, and the faculty adviser. It outlines the end-to-end flow—from the initial submission of payment proofs or student records to the final verification and issuance of digital receipts—highlighting how each role communicates through the platform to ensure an organized, traceable, and efficient administrative process for the student government

Fig. 13 Sequence Diagram

Activity Diagram

Figure 14 presents the Activity Diagram of the SSG Office Assistant App, showing the dynamic workflow among students, SSG officers, faculty advisers, and the system. It outlines the operational path from user authentication to the submission and review of payment proofs and administrative records, proceeding through verification stages until the final issuance of a digital receipt and ledger update. The diagram also highlights the search and retrieval process, where students can check their individual financial standing and officers can access the centralized database to ensure organizational transparency and accountability.

Analysis

Cost and Benefit Analysis

Cost and Benefit Analysis The development of the SSG Office Assistant App is highly cost-effective, leveraging open-source technologies like React, Vite, and Firebase. By utilizing existing university resources and student-led development, the project

eliminates the need for expensive commercial licenses or external developers. This approach aligns with Cebu Technological University – Daanbantayan Campus's goals for digital transformation and administrative transparency without imposing an immediate financial burden.

However, long-term institutional adoption may introduce incremental costs. As the system scales to handle a larger volume of student records and high-resolution media, cloud usage may exceed Firebase's free tier, requiring a transition to a paid model. Additionally, permanent public deployment will necessitate modest investments in web hosting and domain registration. Despite these potential scaling costs, the system remains a sustainable investment, providing significant value in organizational efficiency and secure, paperless record-keeping.

Cost Recovery Scheme

While the project was developed using cost-free, open-source tools, a formal cost recovery scheme is not immediately required. To ensure long-term sustainability, provisions for maintenance and feature enhancements can be supported through Cebu Technological University's existing IT infrastructure or internal innovation grants. Additional funding may be pursued through government-academic partnerships or programs focused on digital educational technology. By integrating routine maintenance and faculty-supervised updates into ongoing research initiatives, the SSG Office Assistant App remains a viable, self-sustaining digital infrastructure for the Daanbantayan Campus.

Risk Assessment/ Analysis

The proponents conducted a risk assessment to identify potential challenges during the development and implementation of the SSG Office Assistant App. The following risks were noted, along with corresponding mitigation strategies to ensure the system's success.

Software Bugs. Technical errors within the application code could disrupt administrative workflows or real-time data synchronization. This will be mitigated by the proponents through thorough unit, integration, and system testing within the Firebase environment to ensure all functional modules are stable before deployment.

User Resistance. Students or SSG officers may be hesitant to transition from traditional paper-based methods to a digital dashboard. This will be addressed by the proponents through targeted orientation programs, training sessions, and the implementation of a user-friendly, intuitive interface designed specifically for the campus community.

Content Quality Issues. Inaccurate or incomplete payment proofs and record uploads may affect the reliability of the centralized ledger. To manage this, the

proponents have implemented Admin-level approval workflows, allowing authorized officers to verify the authenticity and quality of every submission before it is archived.

Internet Connectivity Limitations. Some users may experience limited access to the platform due to varying internet stability on school grounds. The proponents have designed the system to be highly responsive and lightweight, ensuring that it remains accessible across different devices and functions effectively even under lower bandwidth conditions.

Overall, the proponents have ensured that the project carries minimal financial risk while providing significant operational and institutional benefits. The development strategy emphasizes maintainability, scalability, and stakeholder involvement, ensuring the SSG Office Assistant App remains a sustainable and adaptable tool aligned with the long-term digital goals of the student government.

Design

Output and User-Interface Design

The SSG Office Assistant App is designed to provide a streamlined, user-friendly environment tailored to the specific roles of students, SSG officers, and faculty advisers. Its interface focuses on clarity, administrative efficiency, and secure data handling,

supporting core functions such as payment proof submission, financial tracking, record verification, and automated notifications. By offering role-based access and intuitive navigation, the platform enhances the Cebu Technological University – Daanbantayan Campus's organizational workflow, ensuring organized, reliable, and transparent documentation for the entire student body.

Form

The SSG Office Assistant App incorporates a series of role-based forms to guide user interactions and maintain administrative efficiency. Students utilize the Payment & Record Submission Form, which includes specific fields for student name, ID number, fee category (e.g., membership, event fee), amount, and a digital upload of the proof of payment (image or PDF). SSG Officers use the Officer Verification & Ledger Interface to review these submissions, providing real-time feedback, approving entries, or flagging them for clarification while maintaining a digital audit trail. Faculty Advisers manage high-level validation through the Adviser Oversight Form, ensuring that fiscal reports and administrative records are accurate before final archiving.

Additional system modules include Firebase-driven Login & Registration for secure authentication, Profile Management for updating user contact details, and the Record Retrieval Interface for students to check their individual financial standing. These forms ensure that every transaction is structured, traceable, and strictly role-driven.

Reports

The SSG Office Assistant App incorporates advanced reporting features designed primarily for administrative oversight, financial transparency, and data-driven decision-making. System Usage Reports track active users, login frequency, and dashboard engagement to monitor how effectively students and officers are utilizing the platform. Notification Engagement Reports analyze the reach of the Messenger API Gateway, highlighting the success of broadcasted announcements and fine alerts.

Fiscal Statistics Reports provide a comprehensive summary of submissions categorized by fee type (e.g., membership, insurance, school events), department, and year level, while Administrative Audit Reports track the verification speed and productivity of SSG officers. Finally, Financial Transparency Summaries showcase real-time collection totals and budget allocations, ensuring that all stakeholders within the Cebu Technological University – Daanbantayan Campus have access to high-level organizational insights.

Data Design

The SSG Office Assistant App incorporates advanced reporting features designed primarily for administrative oversight, financial transparency, and data-driven decision-making. System Usage Reports track active users, login frequency, and dashboard engagement to monitor how effectively students and officers are utilizing the platform. Notification Engagement Reports analyze the reach of the Messenger API Gateway, highlighting the success of broadcasted announcements and fine alerts.

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Entity Relationship Diagram

Figure 15 presents the Entity Relationship Diagram (ERD) of the SSG Office Assistant App, illustrating the logical data structure designed for implementation within Firebase Firestore. The diagram centers on the USER entity, which encompasses all system users—students, SSG officers (Treasurer, Secretary, Legislative, Budget Sector, Creative Team, etc.), and faculty advisers—with a role attribute enabling role-based access control. Each USER belongs to a DEPARTMENT (e.g., BSIT, BEED) to support department-specific filtering.

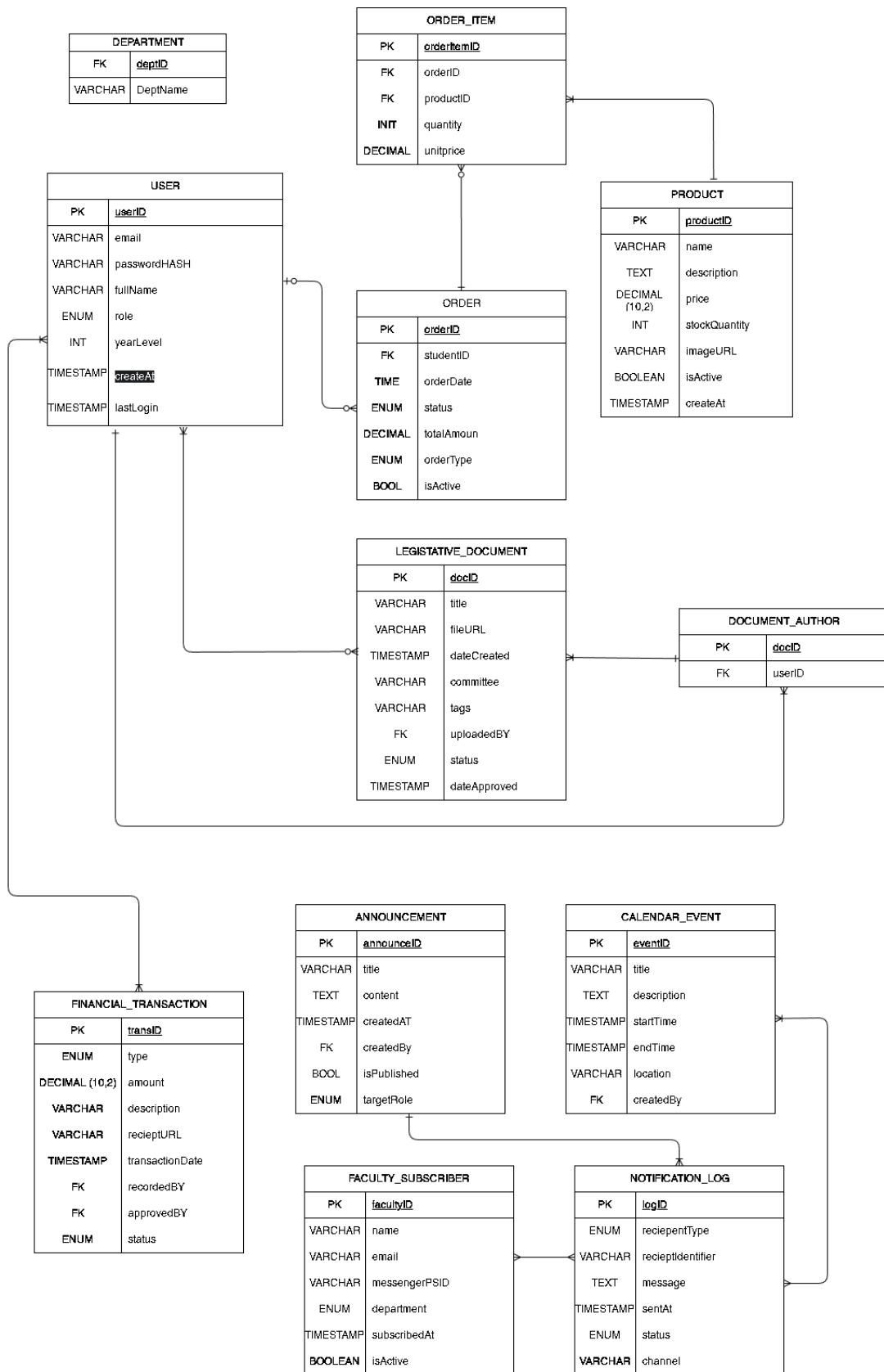


Figure 15. Entity Relation Diagram of The SSG Office Assistant App

Core operational entities include PRODUCT (inventory items), ORDER and ORDER_ITEM (student purchases and order tracking), and FINANCIAL_TRANSACTION (income/expense records with receipt attachments). Legislative documents are managed through the LEGISLATIVE_DOCUMENT entity, linked to authors via the associative DOCUMENT_AUTHOR table to support multiple authors per resolution.

For communication and campus engagement, the diagram includes ANNOUNCEMENT and CALENDAR_EVENT entities, both created by Creative Team members. FACULTY_SUBSCRIBER stores faculty members who receive automated Messenger notifications. NOTIFICATION_LOG tracks all broadcasts sent via the Messenger API, linking each log entry to the source (announcement or event) and the recipient (student or faculty). Overall, the ERD demonstrates a normalized, efficient structure that supports secure storage, real-time synchronization (via Firestore's document model), and strict role-based access control for transparent student government operations.

Data Dictionary

The SSG Office Assistant App utilizes a comprehensive Data Dictionary to serve as a technical blueprint for the platform's Firebase Firestore architecture. This structured reference ensures that every entity—from student payment records to administrative

folders—maintains strict data consistency and logical flow, facilitating seamless integration between the React frontend and the Messenger API Gateway

Table 3. Department Data Dictionary

Name	Data Type	Size	Constraints	Description
deptID	INT	10	PRIMARY KEY AUTO INCREMENT	Unique department identifier
deptName	VARCHAR	50	NOT NULL, UNIQUE	Department name

Table 4. User Data Dictionary

Name	Data Type	Size	Constraints	Description
userId	VARCHAR	28	PRIMARY KEY	User Unique Identifier
email	VARCHAR	100	NOT NULL, UNIQUE	Login Email Address
fullName	VARCHAR	100	NOT NULL	User's Full Name
role	ENUM	-	NOT NULL	GeneralStudent, Legislative, HouseRep, Governor,Budget, SystemAdmin, Adviser
deptID	INT	10	FOREIGN KEY – DEPARTMENT (deptID), Nullable	Department affiliation (null for admins)
profilePhotoUrl	VARCHAR	255	NULLABLE	Firebase Storage URL for profile picture
createdAt	TIMESTAMP	-	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp
lastLogin	TIMESTAMP	-	NULLABLE	Last login timestamp

Table 4. Product Data Dictionary

Name	Data Type	Size	Constraints	Description
productid	INT	10	PRIMARY KEY , AUTO INCREMENT	Unique Product Identifier
Name	VARCHAR	100	NOT NULL	Product Name
Description	TEXT	-	NULLABLE	Product Description
Price	DECIMAL	10,2	NOT NULL	Selling Price PHP
stockQuantity	INT	-	NOT NULL, DEFAULT 0	Current Inventory Count
imageUrl	VARCHAR	255	NULLABLE	Product Image Store URL
isActive	BOOLEAN	-	DEFAULT TRUE	Product Availability
createdAt	TIMESTAMP	-	DEFAULT CURRENT_TIMESTAMP	Product Creation timestamp

Table 5 Order Data Dictionary

Name	Data Type	Size	Constraints	Description
orderID	INT	10	PRIMARY KEY, AUTO INCREMENT	Unique order Number
studentID	VARCHAR	28	FOREIGN KEY – USER(userID),NOT NULL	Student who place his order
orderDate	TIMESTAMP	-	DEFAULT CURRENT_TIMESTAMP	Date and time of order
Status	ENUM	-	NOT NULL. DEFAULT 'pending'	Pending ,processing,complete, cancelled
totalAmount	DECIMAL	10 , 2	NOT NULL	Denormalized sum of order items
department Note	VARCHAR	255	NULLABLE	Optional note for department filtering

Table 6. Order Item Data Dictionary

Name	Data Type	Size	Constraints	Description
orderItemID	INT	-	PRIMARY KEY, AUTO INCREMENT	Unique line item identifier
orderID	INT	-	FOREIGN KEY – ORDER (orderID),NOT NULL	Parent Order
productid	INT	-	NOT NULL, CHECK(>0)	Product being ordered
Quantity	INT	-	NOT NULL	Quantity Ordered
unitprice	DECIMAL	10,2	NOT NULL	Snapshot of price at order time

Table 7. Financial Transaction Data Dictionary

Name	Data Type	Size	Constraints	Description
transID	INT	10	PRIMARY KEY, AUTO_INCREMENT	Unique Transaction identifier
type	ENUM	-	NOT NULL	Income or expense
amount	DECIMAL	10, 2	NOT NULL, CHECK(>0)	Transaction amount in PHP
Description	VARCHAR	200	NOT NULL	Purposes or details
recieptURL	VARCHAR	255	NULLABLE	Firebase Storage URL for receipt image/PDF
transactionDate	TIMESTAMP	-	DEFAULT CURRENT_TIMESTAMP	Date of transaction
recordedBy	VARCHAR	28	FOREGIN KEY – USER (userID), NOT NULL	Budget Sector or President who recorded
approvedBy	VARCHAR	28	FOREIGN KEY – USER (userID), NULLABLE	President who approved
status	ENUM	-	DEFAULT 'pending'	Pending,approved,rejected

Table 8. Legislative Document

Name	Data Type	Size	Constraints	Description
docID	INT	20	PRIMARY KEY , AUTO_INCREMENT	Unique Document Identifier
Title	VARCHAR	150	NOT NULL	Resolution or Document Title
fileURL	VARCHAR	255	NOT NULL	Storage URL for PDF
dateCreated	TIMESTAMP	-	DEFAULT CURRENT_TIMESTAMP	Document Creation date
Committee	VARCHAR	100	NULLABLE	Committee name
Tags	VARCHAR	255	NULLABLE	Comma- seperated keyword for search
uploadedBy	VARCHAR	28	FOREIGN KEY - US	Legislative member who uploaded

Table 9 Document Author Data Dictionary

Name	Data Type	Size	Constraints	Description
docID	INT	20	FOREIGN KEY – LEGISLATIVE_DOCUMENT(docID),PRIMARY KEY	Document identifier
userID	VARCHAR	28	FOREIGN KEY – USER (userID),PRIMARY KEY	Author (officer) Identifier

Table 10. Announcement Data Dictionary

Name	Data Type	Size	Constraints	Description
announced	INT	10	PRIMARY KEY, AUTO INCREMENT	Unique announcement identifier
Title	VARCHAR	100	NOT NULL	Announcement headline

Content	TEXT	-	NOT NULL	Full announcement text
createdAt	TIMESTAMP	-	DEFAULT CURRENT_TIMESTAMP	Publication Timestamp
createdBy	VARCHAR	28	FOREIGN KEY – USER (userID), NOT NULL	Creative Team member who created
isPublished	BOOLEAN	-	DEFAULT TRUE	Whether visible to students
targetRole	ENUM	-	NULLABLE	Restrict to specific role (e.g General Student)

Table 11. Calendar Event Data Dictionary

Name	Data Type	Size	Constraints	Description
eventide	INT	20	PRIMARY KEY, AUTO_INCREMENT	Unique event identifier
Title	VARCHAR	100	NOT NULL	Event name
Description	TEXT	-	NULLABLE	Event Details

startTime	TIMESTAMP	-	NOT NULL	Event start date/time
endTime	TIMESTAMP	-	NOT NULL	Event end date/time
Location	VARCHAR	150	NULLABLE	Venue or online Link
createdBy	VARCHAR	28	FOREIGN KEY – USER(userID), NOT NULL	Multimedia Team Member who created

Table 12 Faculty Subscriber Data Dictionary

Name	Data Type	Size	Constraints	Description
facultyID	INT	10	PRIMARY KEY, AUTO_INCREMENT	Unique faculty identifier
name	VARCHAR	100	NOT NULL	Faculty member's full name
Email	VARCHAR	100	NOT NULL, UNIQUE	Email address for contract
messengerPSID	VARCHAR	100	NOT NULL, UNIQUE	Facebook Page Scoped

				ID for Messenger API
Department	VARCHAR	50	NULLABLE	Academic department assignment
subscribedAt	TIMESTAMP	-	DEFAULT CURRENT_TIMESTAMP	When subscription started
isActive	BOOLEAN	-	DEFAULT TRUE	Whether still receiving notifications

Table 13 Notification Log Data Dictionary

Name	Data Type	Size	Constraints	Description
logId	INT	-	PRIMARY KEY, AUTO INCREMENT	Unique Log Entry
sourceType	ENUM	-	NOT NULL	'announcement' or 'calender event'

sourceID	INT	20	NOT NULL	announced or eventID
recipientType	ENUM	-	NOT NULL	'student' or 'faculty'
recipientIdentifier	VARCHAR	100	NOT NULL	UserID for student or faculty
message	TEXT	-	NOT NULL	Content of notification sent
sentAt	TIMESTAMP	-	DEFAULT CURRENT_TIMESTAMP	When notification was sent
status	ENUM	-	DEFAULT 'success'	Success,failed,pending
channel	VARCHAR	50	DEFAULT 'Messenger'	Deliver Channel

Table 14. Audit Trail Data Dictionary

Name	Data Type	Size	Constraints	Description
auditID	INT	-	PRIMARY KEY , AUTO INCREMENT	Unique audit entry
userID	VARCHAR	28	FOREIGN KEY – USER(userID), NOT NULL	User who performed action
Action	VARCHAR	50	NOT NULL	e.g "CREATE, UPDATE, DELETE,APPROVE "
entityType	VARCHAR	50	NOT NULL	Affected Table
entityID	VARCHAR	50	NOT NULL	Primary key value of affected record
oldValues	JSON	-	NULLABLE	Snapshot before change
newValues	JSON	-	NULLABLE	Snapshot after change
timestamp	TIMESTAMP	-	DEFAULT CURRENT_TIMESTAMP	When Action occurred

System Architecture

Network Model

The SSG Office Assistant App (Figure 16) utilizes a client-server network model to centralize student government operations at CTU-Daanbantayan Campus. In this architecture, students and officers access the dashboard via web browsers on their respective devices (the clients), while the Firebase ecosystem serves as the central server infrastructure. This model ensures that all user accounts, financial ledgers, and administrative records are hosted in a single, secure location.

Fig 16. Client Server Network Model

Network Topology

The SSG Office Assistant App (Figure 17) is architecturally structured using a star network topology, optimized for the specific operational environment of the CTU-Daanbantayan Campus. In this configuration, all client endpoints—including campus desktops, personal laptops, and mobile devices used by the student body—connect directly to the central Firebase server infrastructure.

Fig 17. Star Network Topology

Security

Security in the SSG Office Assistant App is implemented through a multi-layered approach designed to ensure the confidentiality, integrity, and availability of student records and organizational funds. A primary mechanism is Firebase Authentication coupled with Role-Based Access Control (RBAC), which strictly restricts dashboard functionality based on user privileges—limiting Students to their own data while granting Treasurers, Secretaries, and Faculty Advisers specific administrative oversight. Uploaded proofs of payment and official documents are stored in encrypted form within Firebase Cloud Storage, preventing unauthorized access and safeguarding sensitive financial information.

Administrative records and financial validations are regulated through an Officer Verification Workflow, ensuring that only vetted entries are added to the permanent ledger. The system utilizes restricted read-only interfaces for student users, preventing the unauthorized modification or distribution of verified organizational records. To ensure high-level system resilience, regular data backups and comprehensive Audit Trails are integrated, ensuring that all actions are traceable for accountability and that the CTU-Daanbantayan Campus's digital ledger remains a trustworthy environment.

Development

Software Specification

The development of the SSG Office Assistant App at Cebu Technological University – Daanbantayan Campus is predicated on a modular architectural framework designed to synergize administrative workflows among students, student government officers, and faculty advisers. The system incorporates a robust suite of functional modules, starting with secure User Registration and Authentication and a strict Role-Based Access Control (RBAC) mechanism to maintain data confidentiality and ensure that system functionalities are restricted according to user privileges.

To optimize operational efficiency, the platform integrates a Record Submission and Verification module for structured fiscal workflows, complemented by a Document Management system that virtualizes physical office archives for secure digital retrieval. Furthermore, the architecture includes a Notification and Alerts engine to facilitate real-time stakeholder communication via the Messenger API, a Reports and Analytics module for continuous system monitoring, and a comprehensive Audit Log to reinforce organizational transparency. Beyond these functional capabilities, the development prioritizes critical non-functional requirements—specifically security, scalability, usability, and long-term data integrity—to ensure the platform’s reliability and resilience within the university’s digital ecosystem.

Hardware Specification

The technical foundation for the development of the SSG Office Assistant App is predicated on a high-performance computing environment, necessitating workstations equipped with Windows 10 or higher, a minimum of 8GB RAM, 128GB SSD storage, and an 11th Gen Intel Core i5 processor or its equivalent. These specifications are critical to facilitate the fluid operation of integrated development tools, local server environments, and database management protocols required for the Firebase and React ecosystem. On the client side, the platform is engineered for universal accessibility, supporting engagement via modern web browsers on a diverse range of devices, including desktops, laptops, and mobile terminals.

Program Specification

The functional architecture of the SSG Office Assistant App is comprised of a cohesive set of core modules designed to facilitate secure and efficient administrative operations within the CTU-Daanbantayan campus. Central to this framework is User Management, which orchestrates registration, authentication, and the enforcement of role-based access control to maintain system integrity. This is bolstered by an Admin Approval and Moderation engine, which serves as a validation layer for user accounts, record submissions, and access requests.

The platform utilizes a Research/Record Upload module to provide a structured environment for legislative and organizational data entry, while the Access Request

feature permits the regulated retrieval of restricted documentation. To ensure long-term utility, the system incorporates advanced Search and Retrieval capabilities for efficient information discovery, alongside a Reporting and Analytics module that empowers administrators with data-driven insights for institutional oversight. Collectively, these modules ensure a streamlined, secure workflow that optimizes the dissemination of outputs for all university stakeholders.

Front End

The front-end environment for the SSG Office Assistant App is composed of a synergistic suite of web technologies designed to optimize the user interface and client-side interactions. HTML (Hypertext Markup Language) serves as the foundational structural layer, defining the organization of complex web content such as fiscal dashboards, administrative forms, and student record listings. To ensure a professional and responsive user experience, CSS (Cascading Style Sheets) is utilized for visual customization, managing typography, layout, and color schemes to maintain a consistent aesthetic across diverse device profiles. Furthermore, JavaScript—often leveraged within the project's React and Vite ecosystem—introduces essential dynamic behavior, enabling advanced features like real-time search filtering, asynchronous content loading, and rigorous client-side input validation. Collectively, these technologies provide the platform with a seamless and high-performance interface tailored to the specific administrative needs of the campus community.

SSG Office Assistant App is further augmented by specialized React libraries designed to enhance interface interactivity and administrative utility. To provide a

polished and intuitive user experience, Framer Motion is integrated to handle fluid transitions and layout changes within the dashboard. For office-centric functionalities, the platform employs TanStack Table for sophisticated data management of student ledgers and React-Hook-Form for robust validation of fiscal submissions.

Back End

The back-end environment of the SSG Office Assistant App is engineered using a serverless architecture to manage core processes such as user authentication, fiscal submission workflows, and automated report generation. Cloud Firestore serves as the primary NoSQL relational database management system, providing high-performance storage and retrieval of administrative documents, student profiles, and real-time system logs. Unlike traditional local deployments, the system utilizes Firebase Authentication for secure identity management and Cloud Storage for the encrypted archival of digital receipts and official records. This architecture inherently supports role-based access control (RBAC) and data integrity through Firestore Security Rules, ensuring seamless performance and consistent service delivery for multiple concurrent users across the campus network.

Deployment Diagram

The deployment architecture of the SSG Office Assistant App (Figure 18) illustrates the distribution of the platform across cloud-based network and hardware resources. Moving beyond traditional local stacks, the system is deployed on the Firebase cloud infrastructure, hosting a React and Vite frontend and connecting to a

Cloud Firestore NoSQL database with automated redundancy and backups. Students access the platform to submit fiscal records, while SSG Officers and Faculty Advisers conduct administrative reviews and generate reports, all through secure web browsers. The frontend features a robust administrative dashboard and operates over HTTPS for encrypted data transfer. Deployment is supported by a CI/CD pipeline (such as GitHub Actions or Firebase Hosting), ensuring automated, reliable, and scalable updates to the live system to maintain continuous service delivery at CTU-Daanbantayan.

Fig 18. Supreme Student Government App Deployment Diagram

Test Plan

To ensure the SSG Office Assistant App is reliable, secure, and user-friendly, a structured multi-phase testing strategy will be implemented at Cebu Technological University – Daanbantayan Campus. This plan validates individual functional modules—such as User Management and Record Submission—as well as integrated system performance. The testing phases align with the modular system specifications and include unit testing, integration testing, system (alpha) testing, and acceptance testing, following best practices for web-based administrative software development. Each phase is meticulously designed to identify and address defects early in the development lifecycle, ensuring the platform meets both functional and usability requirements prior to institutional deployment.

Table 15. Test Phase

Phase	Description
Unit Testing	Validates the logic of individual functions and components
Integration Testing	Ensures seamless data flow between interconnected modules
System (Alpha) Testing	Evaluates the complete integrated system in a controlled environment
Acceptance Testing	Validates that the system meets user requirements and institutional goals

Testing

The testing phase for the SSG Office Assistant App is a critical lifecycle stage designed to verify that each architectural component functions correctly both independently and as a unified platform. By employing a sequential rigorous testing strategy, the development team ensures that the system maintains high performance, reliability, and institutional trust. This sequence follows a logical progression from code-level validation to final user satisfaction benchmarks.

Unit Testing

The Unit Testing phase for the SSG Office Assistant App focuses on the granular verification of individual functional modules in isolation. By isolating these units, the development process ensures that the fundamental logic of the React components and Firebase functions is robust before they are integrated into the larger system. Each unit is subjected to a rigorous battery of inputs including edge cases and invalid data to confirm accurate outputs and comprehensive error handling.

Table 16. Unit Testing Breakdown

Module Unit	Testing Objective
User & Identity	Ensure secure identity management and correct permission mapping for students and officers
Record & Document	Validate the integrity of fiscal submissions and the reliability of the verification workflow
Search & Discovery	Confirm than the system retrieves the correct records efficiently across large datasets
Feedback & Alerts	Ensure transparent communication between the SSG office and the student Body

Integration Testing

The Integration Testing phase for the SSG Office Assistant App evaluates the seamless interoperability between discrete modules, ensuring that data flows accurately across the system's architectural boundaries. This phase focuses on the critical exchange between the React-based frontend and the Firebase backend, specifically verifying that user authentication tokens correctly authorize access to the Cloud Firestore database and Cloud Storage buckets. Key integration points include the synchronization of the Record Submission module with the Notification and Alerts engine, ensuring that when an officer updates a payment status, the change triggers an automated alert via the Messenger API Gateway.

System/Alpha Testing

The System (Alpha) Testing phase for the SSG Office Assistant App involves comprehensive internal end-to-end evaluations conducted by the development team in collaboration with select faculty and staff members. During this stage, the platform is subjected to realistic operational scenarios, such as a student uploading a proof of payment, an SSG Officer validating the transaction, and a Faculty Adviser reviewing the consolidated financial reports for transparency. The testing team critically examines system performance across a variety of hardware, ensuring responsiveness on mobile

devices and interface consistency throughout the administrative dashboard. Key technical features including the behavior of the digital document viewer, the efficacy of Role-Based Access Control (RBAC) security functions, and the reliability of automated Firebase backup processes are rigorously audited to identify and rectify potential bottlenecks or vulnerabilities prior to the platform's official public release.

Acceptance Testing

The Acceptance Testing phase represents the final validation stage for the SSG Office Assistant App and is conducted following the system's successful oral defense. During this phase, a live pilot implementation is initiated within the Cebu Technological University – Daanbantayan Campus environment to evaluate the platform under real-world conditions. Participants execute core administrative tasks such as authenticating accounts, submitting fiscal records, performing data queries, and reviewing organizational documents to provide qualitative feedback through structured surveys or interviews.

ETHICAL CONSIDERATIONS

The development of the SSG Office Assistant App is an academic undertaking conducted as a primary requirement for the Bachelor of Science in Information Technology program at Cebu Technological University – Daanbantayan Campus.

The research is dedicated to the design and evaluation of a secure, structured digital repository system intended to modernize the preservation, validation, and retrieval of organizational records and institutional materials. To ensure the integrity of the process and the protection of the campus community, the study adheres to a rigorous ethical protocol centered on Voluntary Participation and Informed Consent, ensuring that all involvement is consensual and that participants are fully briefed on the system's objectives.

1. Voluntary Participation – All respondents will participate of their own free will. No participant will be pressured, coerced, or unduly influenced to join or continue in the study.

2. Informed Consent – Before participation, respondents will be briefed about the study's objectives, scope, and procedures. Each participant will sign a consent form to confirm their understanding and agreement to take part in the study.

3. **Right to Withdraw** – Respondents have the absolute right to withdraw or refuse to answer specific questions at any stage of the study without facing any form of penalty or discrimination.

4. **Confidentiality and Privacy** – All collected data will be treated with the utmost confidentiality. Respondent identities will remain anonymous, and all responses will be used solely for academic and evaluative purposes. No personally identifiable information will be shared outside the research team or disclosed to unauthorized individuals.

5. **Purpose and Academic Boundaries** – This study is purely academic and does not involve any commercial, promotional, or financial transactions.

The intent is to gather insights that will contribute to the Digital, evaluation, and improvement of the proposed SSG Assistant App; A Digital Management System for Student Government Operation for the benefit of the university community.

By maintaining these ethical boundaries and ensuring the Right to Withdraw without penalty, the researchers affirm a commitment to safeguarding the dignity and well-being of the university community while delivering a platform that contributes to the digital transformation of the institution.

DEFINITION OF TERMS

Agile Model. Iterative software development process using short cycles and user feedback.

Audit Trail. Chronological log of all user actions and data changes for accountability.

Centralized Database. Single Firestore repository unifying all SSG data to eliminate silos.

Firebase. Serverless Google platform providing authentication, real-time database, and storage.

Firebase Authentication. Service verifying user login and enforcing role-based access.

Fragmented System. Current disconnected SSG operations—paper files, separate spreadsheets, isolated chat groups.

Information Silo. Communication gap where announcements remain trapped in limited groups, excluding key stakeholders.

Messenger API. Facebook's interface enabling automated chatbot announcements to faculty's Messenger.

Role-Based Access Control (RBAC). Security mechanism limiting features and data based on predefined user roles (e.g., Budget Sector, Student).

Single Page Application (SPA). Web app updating dynamically without full reloads.

SSG Office Assistant App. Central digital platform integrating member, financial, document, and communication functions.

Vite. Frontend build tool for fast development and optimized loading.

Webhook Integration. Event-driven trigger that automates Messenger notifications from database changes.

Zero-Trust Environment. Security architecture requiring strict verification for every request.

Legislative Document. Official resolution with metadata (date, authors, tags) for digital archiving.

JSON Web Token (JWT) Custom Claims. Encrypted role attributes attached to user login tokens

React. JavaScript library for building modular, component-based interfaces.

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